Coupled FEM/BEM: Open-Domain Conditions in 2D Electrostatics

Computational Electromagnetics

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The Problem: Parallel-Plate Capacitor in 2D

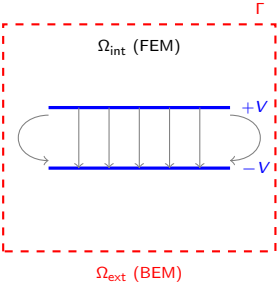
Solve Laplace's equation in the whole plane:

$$\Delta u = 0 \quad \text{in } \mathbb{R}^2 \setminus \{\text{plates}\}$$

with Dirichlet conditions on the plates:

$$u = +V \text{ (top)}, \qquad u = -V \text{ (bottom)}$$

Goal: compute the electric potential *both inside and outside* the capacitor in a real case scenario.



Why Not Pure FEM?

Pure FEM approach

Truncate the domain; impose $u = 0$ on the artificial boundary Γ.

Drawbacks:

- Requires a **very large domain** to minimize the truncation error (Γ must be placed at $\sim 50\, L_y$ from the plates).
- Leads to **large meshes** and long computation times.
- Cannot represent the solution **up to infinity**: introduces an unavoidable approximation.

Key idea

Use **FEM** inside a small domain and **BEM** to handle the exterior exactly — including infinity.

FEM Weak Formulation

Starting from the Laplace problem on a bounded domain Ω:

$$\Delta u = 0 \quad \Longrightarrow \quad \int_\Omega \nabla u \cdot \nabla v \, dx - \underbrace{\int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds}_{\text{boundary term}} = 0 \qquad \forall v \in H^1(\Omega)$$

After discretisation with basis $\{v_i\}_{i=1}^{N_h}$, writing $u \approx \sum_j u_j v_j$ and $\dfrac{\partial u}{\partial n} \approx \sum_j \dfrac{\partial u_j}{\partial n} v_j$:

$$S\,\mathbf{u} \; - \; T\,\frac{\partial \mathbf{u}}{\partial n} = \mathbf{0}$$

where $S_{ij} = \displaystyle\int_\Omega \nabla v_i \cdot \nabla v_j \, dx$ and $T_{ij} = \displaystyle\int_{\partial\Omega} v_i \, v_j \, ds$.

Variable Count — The Key Issue

How many unknowns?

The system provides N_h equations. The unknowns are:

- u_j, $j = 1, \ldots, N_h$ nodal potential values
- $\partial u_j / \partial n$, $j = 1, \ldots, N_h$ normal-derivative coefficients

$$N_h \text{ equations} \quad \text{vs} \quad 2N_h \text{ unknowns} \quad \Rightarrow \quad \textbf{under-determined!}$$

Rescue: Lagrange nodal basis functions satisfy $v_i(x_j) = \delta_{ij}$, so

$$v_i\big|_{\partial\Omega} = 0 \quad \text{for every } \textit{interior} \text{ node } i \quad \Longrightarrow \quad T_{ij} = 0 \text{ if } i \text{ or } j \text{ is interior.}$$

Variable reduction

- Normal-derivative unknowns at *interior* nodes drop out of the system.
- System: N_h equations, $N_h + N_h^\partial$ unknowns. Here BEM provide N_h^∂ equations.

FEM Block Structure

Partitioning nodes into **interior** (I) and **boundary** (B):

$$\underbrace{\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} \end{bmatrix}}_{\text{FEM stiffness}} \begin{bmatrix} u_I \\ u_B \end{bmatrix} - \begin{bmatrix} \mathbf{0} \\ T_{BB} \dfrac{\partial u_B}{\partial n} \end{bmatrix} = \mathbf{0}$$

- Row I: closed — $\partial u / \partial n$ does not appear.
- Row B: **open** — involves the unknown $\partial u_B / \partial n$ (will be supplied by BEM).

BEM — Boundary Integral Equation

Key idea: reformulate the Laplace problem on $\partial\Omega$ only, using the fundamental solution $G(x,y) = \frac{1}{2\pi}\log|x-y|$ in 2D.

Green's third identity $\Rightarrow$ **Boundary Integral Equation** (BIE):

$$\frac{1}{2}\,u(y) = \underbrace{\int_{\partial\Omega} G(x,y)\,\frac{\partial u}{\partial n}(x)\,ds_x}_{\mathcal{V}(\partial u/\partial n)(y)} - \underbrace{\int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x,y)\,u(x)\,ds_x}_{\mathcal{K}(u)(y)} \qquad \forall\, y \in \partial\Omega$$

After collocation discretizations with boundary basis $\{\psi_j\}$:

$$H\,u_B = G\,\frac{\partial u_B}{\partial n} \qquad H = \tfrac{1}{2}M + K, \quad G \text{ symmetric}$$

BEM provides **exactly** N_h^∂ equations relating u_B and $\partial u_B/\partial n$ — the missing relations from FEM.

Coupled System and Plates Conditions

Replace the open FEM boundary rows with the BEM equations:

$$\begin{bmatrix} S_{II} & S_{IB} & 0 \\ S_{BI} & S_{BB} & -T_{BB} \\ 0 & H & -G \end{bmatrix} \begin{bmatrix} u_I \\ u_B \\ \partial u_B / \partial n \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

- **Row 1 (FEM, interior):** governs u inside Ω.
- **Row 2 (FEM, boundary):** couples u_I, u_B with the flux.
- **Row 3 (BEM):** closes the system via integral operators on $\partial\Omega$.

Elimination and Reduced System

Assume G invertible:

$$\frac{\partial u_B}{\partial n} = -G^{-1} H\, u_B$$

Substitute into the FEM boundary row $\Rightarrow$ **reduced system of size** N_h:

$$\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} + \underbrace{T_{BB}\, G^{-1} H}_{\text{BEM correction (dense)}} \end{bmatrix} \begin{bmatrix} u_I \\ u_B \end{bmatrix} = - \begin{bmatrix} S_{ID} \\ S_{BD} \end{bmatrix} u_D$$

- Same size as a standard FEM system ($N_h \times N_h$).
- The BEM term $T_{BB} G^{-1} H$ is **dense** but restricted to boundary DOFs.
- In MATLAB: $G^{-1} H$ computed via the backslash operator.
- After solving: recover $\partial u_B / \partial n$ and the exterior potential via the BEM representation formula.

Setup

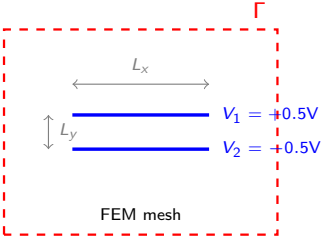
Geometry:

- $L_x = 0.05$ m, $L_y = 0.05$ m ($L_x/L_y = 1$)
- $V_1 = 0.5$ V (top), $V_2 = -0.5$ V (bottom)
- Γ placed at $3\,L_y$ margin on all sides
- $\varepsilon_r = 1$

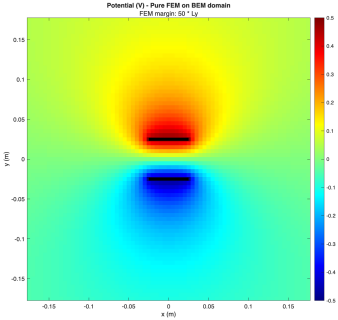
Meshes:

- Coarse: $\delta = 0.005$ m ≈ 5000 nodes

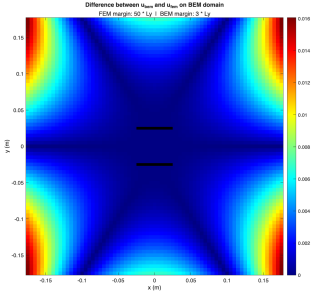
Reference: The reference is a pure FEM method with a Dirichlet condition imposed on Γ placed at $50L_y$



FEM-BEM vs. Pure FEM



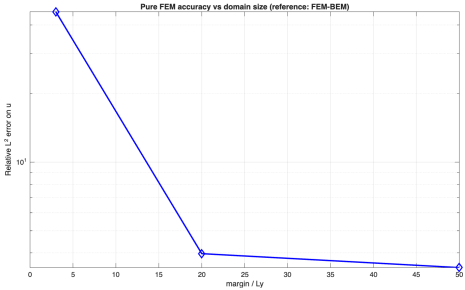
Pure FEM ($20\,L_y$ domain) — reference solution



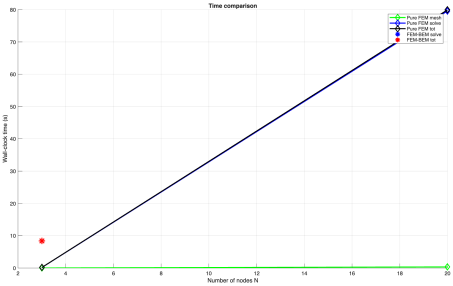
Pointwise difference: FEM vs. FEM-BEM on common domain

$$\varepsilon_{L^2} = \sqrt{\frac{\sum_{i\in\mathcal{N}_c}(u_i^{\text{FEM}} - u_i^{\text{FEM-BEM}})^2}{\sum_{i\in\mathcal{N}_c}(u_i^{\text{FEM}})^2}} \times 100\% \;=\; \mathbf{3.44}\%$$

FEM-BEM vs. Pure FEM — Accuracy & Speed



L^2 error vs. FEM domain size

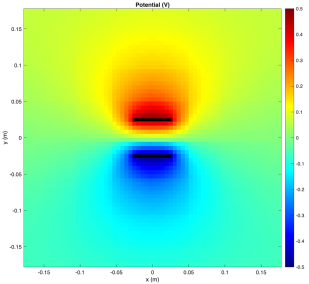


Computational time comparison

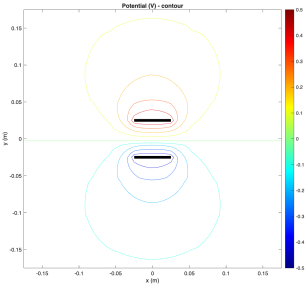
Efficiency gain *on my computer*

- FEM-BEM ($3\,L_y$ domain): **8.52 s**
- Pure FEM ($50\,L_y$ domain): **2800 s** (≈ 47 min)

Electric Potential Distribution FEM-BEM



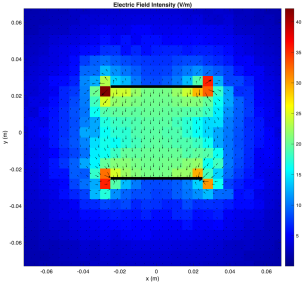
Potential map ($\delta = 0.005$ m)



Equipotential lines

- Uniform field between plates; smooth fringing at edges.

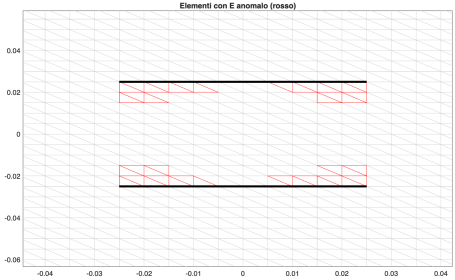
Electric Field FEM-BEM



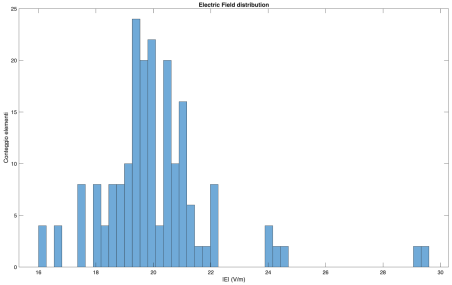
$\delta = 0.005$ m (coarse)

- Central field $\approx (V_1 - V_2)/L_y = 20$ V/m as expected.
- Fine mesh resolves singularity at plate corners (physically expected).
- Coarse mesh smears the edge singularity.

Electric Field — Element Analysis (FEM-BEM)



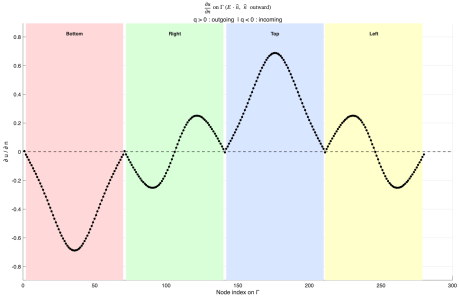
Elements with $|\mathbf{E}| > 21$ V/m



Distribution of $|\mathbf{E}|$ per element

- Anomalous elements localised at plate **corners** only.
- This is a well-known physical singularity of infinitely thin conductors — **not** a numerical artefact.
- The majority of elements have $|\mathbf{E}| \approx 20$ V/m.

Flux Through Γ (FEM-BEM)



$\partial u/\partial n$ on Γ

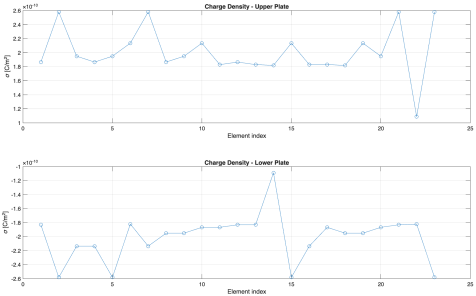
Physical expectation:

- Net flux must vanish (equal and opposite plate charges).

$$Q_{tot} = \oint_\Gamma \frac{\partial u}{\partial n}\, ds \approx -1.79 \times 10^{-13} \approx 0$$

Numerically consistent with zero — coupling condition verified.

Surface Charge Density and Capacitance



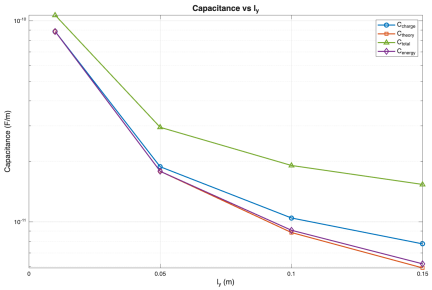
Surface charge density on the plates

Charge method:

$$C = \frac{|Q|}{V_1 - V_2}$$

- $C_{\text{FEM-BEM}} = 9.9811 \times 10^{-12}$ F/m
- $C_{\text{FEM}} = 9.9824 \times 10^{-12}$ F/m

Capacitance different Ly, fixed Lx



Capacitance with different Ly

- **Method comparison:** internal energy method inside the capacitor shows excellent agreement; charge-based method slightly overestimates
- **Full domain energy:** increasing deviation from theory due to energy stored outside the capacitor

Conclusions

- A **FEM-BEM coupled solver** for 2D electrostatics has been implemented in MATLAB, targeting the parallel-plate capacitor problem.
- The coupling exploits:
  1. FEM's accuracy in the bounded interior region.
  2. BEM's ability to represent the exterior field **up to infinity**.
- Validation confirms:
  - Correct potential and field distributions.
  - Capacitance within 0.69% of the reference FEM solution.
  - Gauss's law satisfied numerically.
- **100**$\times$ **speed-up** over a comparable pure FEM simulation.